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Student Number

2016

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Extension 1 Mathematics

General Instructions

- Reading time – 5 minutes
- Working time - 2 hours
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-14 show relevant mathematical reasoning and/or calculations
- Start a new booklet for each question

Total Marks – 70

Section I - Pages ***

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II - Pages ****

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Question	Marks
1 - 10	/10
11	/15
12	/15
13	/15
14	/15
Total	/70

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 40% of the Higher School Certificate Course Assessment

Section I

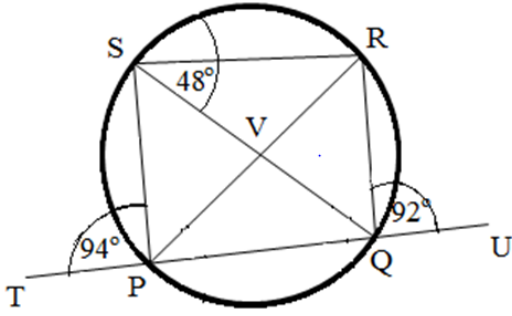
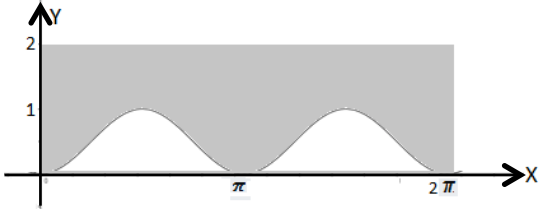
10 marks

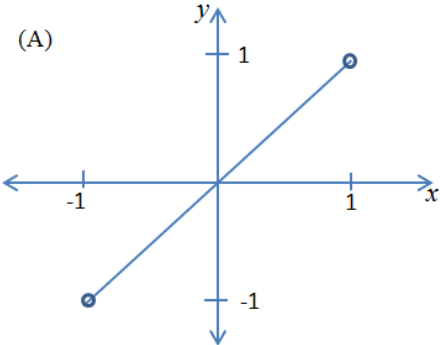
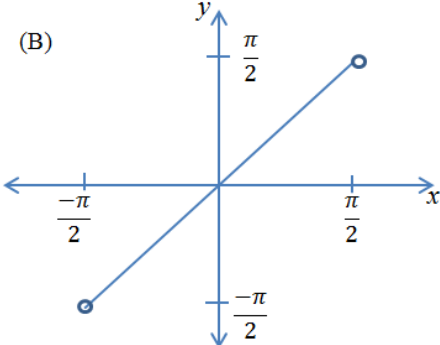
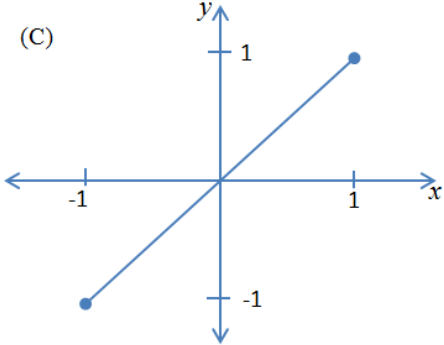
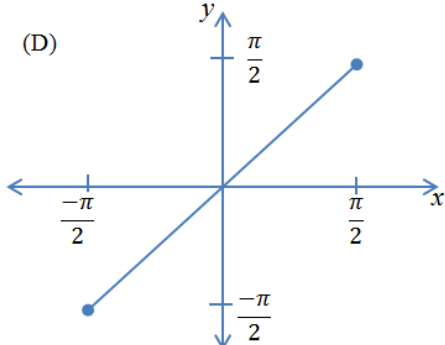
Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1 – 10 (Detach from paper)

1)	In how many ways can 8 students be arranged if the tallest is first and the shortest is last? (A) 8C_6 (B) 6C_6 (C) 8P_6 (D) 6P_6
2)	If $x = 1.6$ is a close root of the equation $x^3 - 4x + 2 = 0$, find a better approximation to two decimal places. (A) 1.68 (B) 13.71 (C) 0.43 (D) 4.96
3)	Evaluate $\sin^{-1}\left(-\frac{1}{2}\right)$ as an exact answer. (A) $\frac{\pi}{6}$ (B) $\frac{5\pi}{6}$ (C) $-\frac{\pi}{6}$ (D) $-\frac{5\pi}{6}$

4)	A cone has a base diameter of 16 cm and a perpendicular height of 12 cm. The angle the side of the cone makes with its base is: (A) 56° (B) 37° (C) 34° (D) 53°
5)	Given $\angle TPS = 94^\circ$, $\angle RQU = 92^\circ$, $\angle QSR = 48^\circ$, find $\angle SVR$  (A) 82° (B) 92° (C) 94° (D) 96°
6)	 The diagram shows a sketch of the curve $y = \sin^2 x$ between $x=0$ and $x= 2\pi$. The shaded area equals: (A) 2π square units (B) 3π square units (C) $2(2\pi - 1)$ square units (D) 4π square units

7)	The equation $\cos(2x) = \operatorname{cosec}\left(x - \frac{\pi}{2}\right)$ $-\pi \leq x \leq \pi$, has how many solutions? (A) 0 (B) 1 (C) 2 (D) 3
8)	Which of the following is the graph of $\sin(\sin^{-1}x)$? (A)  (B)  (C)  (D) 
9)	Using the substitution $u = \sqrt{x}$, $\int \frac{dx}{x+\sqrt{x}}$ can be transformed to: (A) $\int \frac{2du}{u+1}$ (B) $\int \frac{du}{u^2+u}$ (C) $\int \frac{2du}{u^2+u}$ (D) $\frac{1}{2} \int \frac{du}{u^2+u}$

10)	The speed v m/s of a point moving along the x axis is given by $v^2 = 36 + 6x - 2x^2$, where x is in meters. The period and amplitude of the motion are: (A) Period π and amplitude $\frac{9}{2}$ (B) Period 2π and amplitude $\frac{\sqrt{63}}{2}$ (C) Period $\sqrt{2}\pi$ and amplitude $\frac{9}{2}$ (D) Period $\sqrt{2}\pi$ and amplitude $\frac{3}{2}$
11)	A particle undergoes linear acceleration according to the equation $a = (x + 4)^3 \text{ m/s}^2$. Given that the particle commences motion at the origin with a velocity 4 m/s, what is the particle's displacement when $v = 10 \text{ m/s}$, given that $x < 0$? (A) -0.1313 (B) -7.7606 (C) -8.5378 (D) -20.0478

Section II

70 marks

Attempt Questions 11 – 14

Allow about 1 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

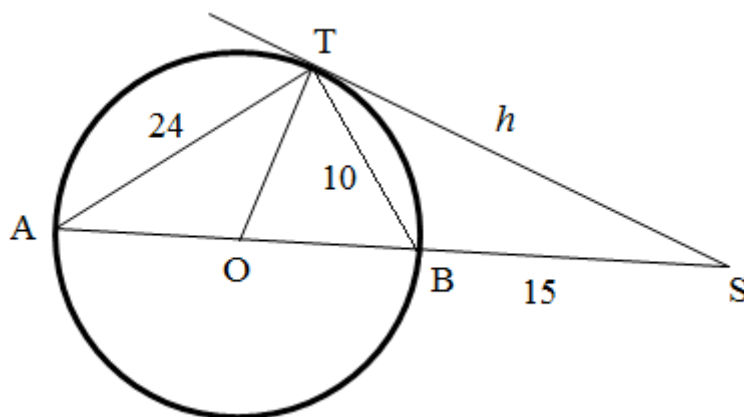
In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

- (a) Solve $\frac{3}{2x-1} < 2$ 3
- (b) Find $\int_0^{\frac{\pi}{2}} \frac{dx}{9+4x^2}$ giving your answer in exact form 3
- (c) Let A be the point $(-2, 5)$ and B be the point $(4, 1)$ 2
Find the coordinates of the point P that divides the interval externally in the ratio 4: 3.
- (d) Evaluate $\int \frac{x \, dx}{\sqrt{x-2}}$ using the substitution $u^2 = x - 2$ 3
- (e) Evaluate $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{x}{3}\right)}{2x}$ 2
- (f) Find $\frac{d}{dx}\left(e^{\sin^{-1}(3x)}\right)$ 2

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) ST is a tangent at T, AT = 24cm, BT = 10cm, BS = 15cm and ST = h cm. O is the centre of the circle. 2



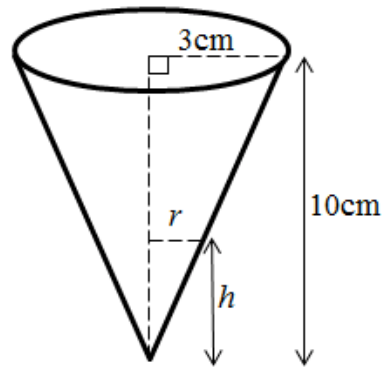
Find the value of h correct to one decimal place

- (b) i) Express $\sqrt{3} \cos \theta - \sin \theta$ in the form $R \cos(\theta + \alpha)$ where $R > 0$ and $0 < \alpha < 2\pi$ 2
- ii) Hence or otherwise solve the equation $\sqrt{3} \cos \theta - \sin \theta = 1$, for $0 \leq \theta \leq 2\pi$ 2
- (c) Solve the equation $x^3 - 21x^2 + 126x - 216 = 0$ given that the three roots form a geometric series 3
- (d) The acceleration of a particle moving in a straight line is given by $\frac{d^2x}{dt^2} = -2e^{-x}$ where x is the displacement (in metres) from the origin. Initially the particle is at the origin and is moving with a velocity of $2ms^{-1}$
- i) Prove that $v = 2e^{\frac{-x}{2}}$ 2
- ii) Find an expression for the displacement x at any time t . 2

Question 12 continues on page 8

Question 12 (continued)

- (e) The diagram shows a conical wheat flue. The flue is being filled at the rate of 2m^3 / minute. The height of wheat at any time, ' t ' minutes, is ' h ' metres, and the radius of the wheat's top surface is ' r ' metres.



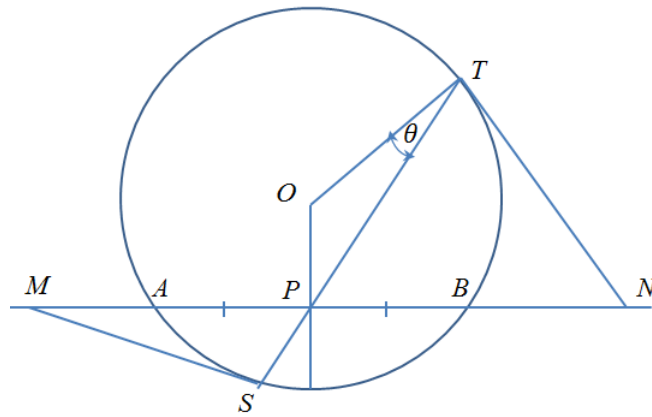
- i) Show that $r = \frac{3h}{10}$ 1
- ii) Find the rate at which the height is increasing when the height of wheat is 8 m 3
- (The volume of a cone is given by $V = \frac{1}{3}\pi r^2 h$)

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) Consider the function $f(x) = 2 \sin^{-1}(x-1)$.
- iii) Find the domain and range of the function. 1
- iv) Sketch the graph of the function. 1
- v) Find the equation of the inverse function. 2
- (b) Four people go to a pizza festival, where four different gourmet pizzas A, B, C and D are offered. Each person chooses a pizza at random to try.
- i) Find the probability that they all choose different pizzas. 2
- ii) Find the probability that exactly two of the people choose pizza A. 2
- (c) i) $T(2at, at^2)$ is a point on the parabola $x^2 = 4ay$. Show that the normal to the parabola at T has equation $x + ty - 2at - at^3 = 0$. 2
- ii) P and Q are points on the parabola $x^2 = 4ay$ with parameter values $t = 1$ and $t = 2$ respectively. Show that the normal to the parabola at P and Q intersect at a point R on the parabola. 2
- (d) The formula for the n th term a_n of the Fibonacci sequence, 3
- 1, 1, 2, 3, 5, 8, 13, 21, 34, ...
- is given by,
- $$a_n = \begin{cases} 1 & \text{for } n = 1 \text{ and } 2 \\ a_{n-2} + a_{n-1} & \text{for } n > 2 \end{cases}$$
- Prove by mathematical induction that,
- $$a_n = \frac{(1+\sqrt{5})^n - (1-\sqrt{5})^n}{\sqrt{5}2^n}$$

Question 14 (15 marks) Use a SEPARATE writing booklet

(a)



In the diagram above, P is the midpoint of the chord AB in the circle with centre O . A second chord ST passes through P , and the tangents at the endpoints meet, AB produced at M and N respectively.

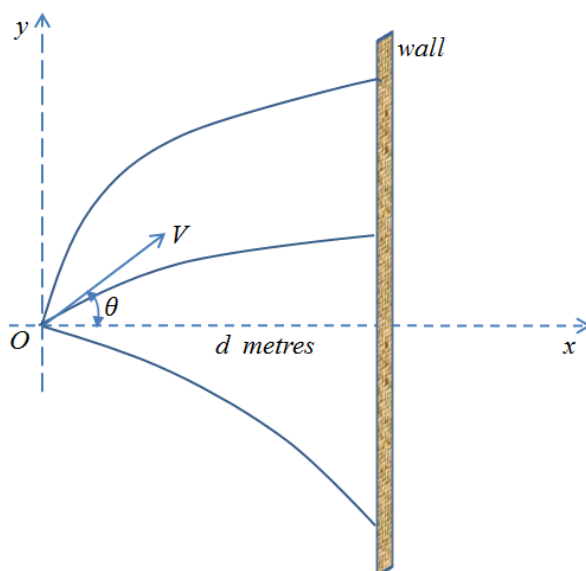
Join OS .

- i) Explain why $OPNT$ is a cyclic quadrilateral. 2
- ii) Explain why $OPSM$ is also cyclic. 1
- iii) Let $\angle OTS = \theta$. Show that $\angle ONP = \angle OMP = \theta$. 2
- iv) Hence, prove that $AM = BN$. 1

Question 14 continues on page 11

Question 14 (continued)

(b)



In the diagram above, a large number of projectiles are fired simultaneously from O , each with the same velocity V m/s, but different angles of projection θ , at a wall d metres from O . The projectiles are fired so that they all lie in the same vertical plane perpendicular to the wall.

You may assume that the equations of motion at time t are given by:

$$x = Vt \cos \theta \quad \text{and} \quad y = -\frac{1}{2}gt^2 + Vt \sin \theta.$$

- i) Using these two equations of motion, prove that the relationship between the height y and time t is: 2

$$4y^2 + 4gt^2y + (g^2t^4 + 4x^2 - 4v^2t^2) = 0.$$

- ii) Show that the first impact at the wall occurs at time $t = \frac{d}{V}$ and that this projectile was fired horizontally. 2

- iii) Hence, find where this projectile hits the wall. 1

- iv) Show that for $t > \frac{d}{V}$, there are *two* impacts at time t , and that the distance between these is: 2

$$2\sqrt{V^2t^2 - d^2}.$$

- v) Given that $V = 10$ m/s and $d = 10$ metres, what are the initial angles of projection of the two projectiles that will strike the wall simultaneously $20\sqrt{3}$ metres apart. 2

End of Examination ☺

Multiple choices

(1) $I = \frac{5}{6P}$ (D)

(2) $f(x) = 3x^2 - 4$
 $f'(1.6) = 3.68$
 $f(1.6) = -0.304$ (A)
 $x_2 = 1.6 - \frac{-0.304}{3.68} = 1.68$

(3) $\sin^{-1}(-\frac{1}{2}) = -\frac{\pi}{6}$ (C)

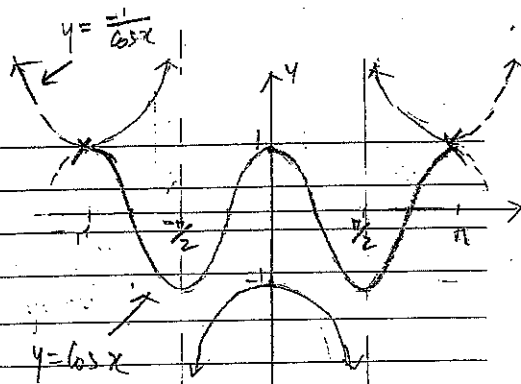
(4) $y = \frac{e^x - 2}{e^x + 2} = \frac{e^x + 2 - 4}{e^x + 2} = 1 - \frac{4}{e^x + 2}$ (B)
 when $x \rightarrow \infty$, $y \rightarrow 1$
 when $x \rightarrow -\infty$, $y \rightarrow 1 - \frac{4}{0+2} = -1$

(5) (A)

(6) $A = 2 \times 2\pi - \int_0^{2\pi} \sin^2 x$
 $= 4\pi - \frac{1}{2} \int_0^{2\pi} (1 - \cos 2x) dx$ (B)
 $= 4\pi - \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{2\pi}$
 $= 3\pi$

$$\textcircled{7} \quad \cos 2x = -\frac{1}{\sin(\frac{\pi}{2}-x)}$$

$$= \frac{-1}{\cos x}$$



from the graph, there are 2 solutions or.

$$\text{solve } \cos 2x = -\frac{1}{\cos x}$$

$$2\cos^2 x - \cos x + 1 = 0$$

$$\Rightarrow x = \pm \pi$$

$$\textcircled{8} \quad \sin(\sin^{-1} x) = x$$

$$\text{domain } -1 \leq x \leq 1$$

$$\text{Range } -1 \leq 1 \leq 1$$

$\textcircled{C}$ ✓

$\textcircled{9}$

$$u = \sqrt{x}$$

$$du = \frac{1}{2\sqrt{x}} dx \Rightarrow dx = 2u du$$

$$\int \frac{dx}{x + \sqrt{x}} = \int \frac{2u du}{u^2 + u} = 2 \int \frac{du}{u+1}$$

$\textcircled{A}$

$\textcircled{10}$

$$\frac{d}{dx} \frac{1}{2} v^2 = \frac{d}{dx} (18 + 3x - x^2)$$

$$= 3 - 2x = 2\left(\frac{3}{2} - x\right)$$

$$n = \sqrt{2} \quad T = \frac{2\pi}{\sqrt{2}} = \sqrt{2}\pi$$

$\textcircled{C}$ ✓

$$x^2 - 3x - 18 = 0$$

$$x = 6$$

$$a = \frac{9}{2}$$

$$(x-6)(x+3) = 0$$

$$x = -3$$

Q11

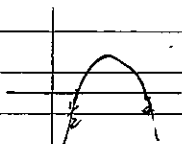
a) $\frac{3}{2x-1} < 2$

$3(2x-1) = 2(2x-1)^2 < 0$

$(2x-1)(3-4x+2) < 0$

$(2x-1)(5-4x) < 0$

$x < \frac{1}{2}$ and $x > \frac{5}{4}$



✓

generally well done

✓

✓

b) $\int_0^{\sqrt{3}/2} \frac{dx}{9+4x^2} = \int_0^{\sqrt{3}/2} \frac{dx}{3^2+(2x)^2}$

$= \frac{1}{6} \left[\tan^{-1} \frac{2x}{3} \right]_0^{\sqrt{3}/2}$

$= \frac{1}{6} \times \frac{\pi}{6} = \frac{\pi}{36}$

✓

✓

✓

generally well done

c) $x = \frac{-4 \times 4 + 3 \times -2}{-1} = 22$

(22, -11)

$y = \frac{-4 \times 1 + 3 \times 5}{-1} = -11$

✓

✓

generally well done

111 Cont.

(d) $u^2 = x-2 \Rightarrow x = u^2+2$

$2u du = dx$

$\int \frac{x dx}{\sqrt{x-2}} = \int \frac{u^2+2}{u} \cdot 2u du$

$= \int 2u^2 + 4 du$

$= \frac{2}{3} u^3 + 4u + C$

$= \frac{2}{3} (x-2)^{3/2} + 4(x-2) + C$

generally well done

✓

✓

few students did not express the answer in terms of x .

✓

e) $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{\frac{x}{3}} \times \frac{1/3}{2} = 1 \times \frac{1}{6}$
 $= \frac{1}{6}$

✓ ✓ Working must be shown correctly to obtain full marks

generally well done

✓ ✓

f) $\frac{d}{dx} e^{\sin^{-1}(3x)} = e^{\sin^{-1}(3x)} \cdot \frac{3}{\sqrt{1-9x^2}}$
 $= \frac{3e^{\sin^{-1}(3x)}}{\sqrt{1-9x^2}}$

generally well done

Q12

a) $R = \sqrt{3+1} = 2$

✓

generally well done

$\alpha = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$

✓

$\therefore \sqrt{3} \cos \theta - \sin \theta = 2 \cos \left(\theta + \frac{\pi}{6} \right)$

$2 \cos \left(\theta + \frac{\pi}{6} \right) = 1$

generally well done

$\theta + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}$

✓

$\therefore \theta = \frac{\pi}{6} \text{ and } \frac{3\pi}{2}$

✓

c) let the 3 roots be $\frac{a}{r}, a$ and ar

$\sum \text{roots} = \frac{a}{r} + a + ar = 21 \quad \text{①}$

Prod. of roots $\frac{a}{r} \cdot a \cdot ar = 216$

$a^3 = 216$

$a = 6 \text{ Sub} \rightarrow \text{②}$

✓ correct value of a

• few students used polynomial division to find the answers.

$\Rightarrow \frac{6}{r} + 6 + 6r = 21$

$6r^2 - 15r + 6 = 0$

$(2r-1)(r-2) = 0$

$\Rightarrow r = \frac{1}{2} \text{ or } 2$

✓ correct working

• few students had no idea how to approach this problem.

$\therefore$ The 3 roots are 3, 6 and 12

✓ correct answers

$$c) \frac{d}{dx} \frac{1}{2} v^2 = -2e^{-x}$$

$$\frac{1}{2} v^2 = 2e^{-x} + C$$

$$\text{when } x=0, v=2$$

$$\Rightarrow C = 2 - 2 = 0$$

$$\therefore \frac{1}{2} v^2 = 2e^{-x}$$

$$v^2 = 4e^{-x}$$

$$v = \pm 2e^{-x/2}$$

$$\text{when } t=0, x=0, v=2\text{m/s} > 0$$

$$\therefore v = 2e^{-x/2}$$

✓ correct expression of $\frac{1}{2} v^2$

✓ correct answer of v^2

✓ final answer with reason

many students assumed that $v = 2e^{-x/2}$ without giving reason.

$$(ii) \frac{dx}{dt} = 2e^{-x/2}$$

$$dt = \frac{1}{2} e^{x/2}$$

$$\Rightarrow t = e^{x/2} + C$$

$$\text{when } t=0, x=0 \Rightarrow C=-1$$

$$\therefore e^{x/2} = t+1$$

$$\text{hence } x = 2\ln(t+1)$$

$$\text{or } = \ln(t+1)^2$$

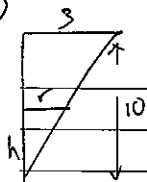
✓ correct expression for t

✓ correct answer

generally well done.

Q12 cont

e)

using similar $\Delta s'$

$$\frac{r}{3} = \frac{h}{10}$$

$$\Rightarrow r = \frac{3h}{10}$$

well done

$$ii) V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi \left(\frac{3h}{10} \right)^2 h$$

$$= \frac{3\pi h^3}{100}$$

$$\frac{dV}{dh} = \frac{9\pi h^2}{100}$$

✓ correct expression for $\frac{dV}{dh}$

many students finding

 $\frac{d}{dh} \frac{1}{3} \pi r^2 h$ withoutsub. $h = \frac{3h}{10}$ before hand. This gave incorrect answer for $\frac{dV}{dh}$.

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} \quad \text{where } \frac{dV}{dt} = 2 \text{ cm}^3/\text{min.}$$

$$= \frac{100}{9\pi h^2} \times 2 = \frac{200}{9\pi h^2}$$

✓ correct expression for $\frac{dh}{dt}$

$$\text{Sub } h = 8$$

$$\Rightarrow \frac{dh}{dt} = \frac{200}{9 \times \pi \times 8^2}$$

$$= \frac{25}{72\pi} \text{ cm}$$

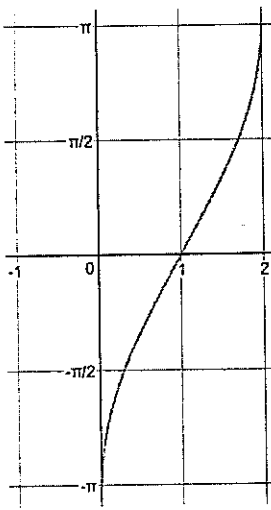
$$= 0.11 \text{ cm}$$

✓ correct substitution & answer.

Some did the long way using

$$\frac{dh}{dt} = \frac{dV}{dr} \times \frac{dr}{dh} + \frac{dV}{dh}$$

and find these derivative functions.

(a)	Consider the function $f(x) = 2\sin^{-1}(x-1)$.							
	i) Find the domain and range of the function.	1						
	<table border="1"> <tr> <th>Criteria</th> <th>Marks</th> </tr> <tr> <td> Provides a correct solution </td> <td>1</td> </tr> </table>	Criteria	Marks	Provides a correct solution	1			
Criteria	Marks							
Provides a correct solution	1							
	<i>Sample answer:</i> Domain: $\{-1 \leq x-1 \leq 1\}$: $\{0 \leq x \leq 2\}$ Range: $\{-\pi \leq y \leq \pi\}$ <i>Mostly done well</i>							
	ii) Sketch the graph of the function.	2						
	<table border="1"> <tr> <th>Criteria</th> <th>Marks</th> </tr> <tr> <td> - Draws a correct sketch </td> <td>2</td> </tr> <tr> <td> - Draws a graph showing the correct shape, or equivalent merit </td> <td>1</td> </tr> </table>	Criteria	Marks	Draws a correct sketch	2	Draws a graph showing the correct shape, or equivalent merit	1	
Criteria	Marks							
Draws a correct sketch	2							
Draws a graph showing the correct shape, or equivalent merit	1							
	<i>Sample solution:</i>  <i>be careful the shape is right.</i> <i>(2, pi) is a horizontal tangent</i>							

	iii) Find the equation of the inverse function.	2						
	<table border="1"> <tr> <th>Criteria</th> <th>Marks</th> </tr> <tr> <td>• Provides a correct solution</td> <td>2</td> </tr> <tr> <td>• Swaps variables and makes some progress towards a correct solution, or equivalent merit.</td> <td>1</td> </tr> </table>	Criteria	Marks	• Provides a correct solution	2	• Swaps variables and makes some progress towards a correct solution, or equivalent merit.	1	
Criteria	Marks							
• Provides a correct solution	2							
• Swaps variables and makes some progress towards a correct solution, or equivalent merit.	1							
	Sample solutions: Swap unknowns $y = 2\sin^{-1}(x-1)$ $\frac{1}{2}y = \sin^{-1}(x-1)$ $\sin\left(\frac{y}{2}\right) = x-1$ $x = 1 + \sin\left(\frac{y}{2}\right)$ hence $f^{-1}(x) = 1 + \sin\left(\frac{x}{2}\right) \text{ for } -\pi \leq x \leq \pi$ <i>well done</i>							
(b)	Four people go to a pizza festival, where four different gourmet pizzas A, B, C and D are offered. Each person chooses a pizza at random to try.							
	i) Find the probability that they all choose different pizzas.	1						
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Criteria	Marks							
• Provides a correct solution	1							
	Sample solution: $P(\text{all different}) = \frac{4!}{4^4}$ $= \frac{3}{32}$							
	ii) Find the probability that exactly two of the people choose pizza A.	2						
	<table border="1"> <tr> <th>Criteria</th> <th>Marks</th> </tr> <tr> <td>• Provides a correct solution</td> <td>2</td> </tr> <tr> <td>• Makes progress towards an answer with at most one option missed or one element unconsidered, or equivalent merit.</td> <td>1</td> </tr> </table>	Criteria	Marks	• Provides a correct solution	2	• Makes progress towards an answer with at most one option missed or one element unconsidered, or equivalent merit.	1	
Criteria	Marks							
• Provides a correct solution	2							
• Makes progress towards an answer with at most one option missed or one element unconsidered, or equivalent merit.	1							
	Sample solution: $P(\text{exactly 2 choose A}) = {}^4C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^2$ $= \frac{27}{128}$ <i>make sure to show working</i>							

(c)	P and Q are points on the parabola $x^2 = 4ay$ with parameter values $t = 1$ and $t = 2$ respectively.							
	i) Given the equation of the normal at T is $x + ty - 2at - at^3 = 0$ (Do not prove). Find the equation of the normal at P and at Q .	1						
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Criteria	Marks							
• Provides a correct solution	1							
	<i>Sample solution:</i> when t is 1 the normal at P: $x + 1y - 2a \cdot 1 - a \cdot 1^3 = 0$ $x + y = 3a$ when t is 2 the normal at Q: $x + 2y - 2a \cdot 2 - a \cdot 2^3 = 0$ $x + 2y = 12a$ <i>Many students confused p & q as variables instead of substituting in 1 & 2</i>							
	ii) Show that the normal to the parabola at P and Q intersect at a point R on the parabola.	2						
	<table border="1"> <tr> <th>Criteria</th> <th>Marks</th> </tr> <tr> <td>• Provides a correct solution</td> <td>2</td> </tr> <tr> <td>• Solve the normal equations simultaneously and attempt to demonstrate solution lies on the parabola, or equivalent merit</td> <td>1</td> </tr> </table>	Criteria	Marks	• Provides a correct solution	2	• Solve the normal equations simultaneously and attempt to demonstrate solution lies on the parabola, or equivalent merit	1	
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• Provides a correct solution	2							
• Solve the normal equations simultaneously and attempt to demonstrate solution lies on the parabola, or equivalent merit	1							
	<i>Sample solution:</i> $x + y = 3a \quad \text{--- 1}$ $x + 2y = 12a \quad \text{--- 2}$ subtracting 2 from 1 $-y = -9a$ $y = 9a$ substituting into 1 $x + 9a = 3a$ $x = -6a$ giving $R(-6a, 9a)$ as the point of intersection $x^2 = 36a^2$ $= 4a(9a)$ $= 4ay$ $\therefore R$ lies on the parabola <i>Some students forget to show the point lies on the parabola</i>							

(d)	i) Prove the following by the process of mathematical induction. $\ln(2) + \ln\left(\frac{3}{2}\right) + \ln\left(\frac{4}{3}\right) + \dots + \ln\left(\frac{n+1}{n}\right) = \ln(n+1)$	3								
	<table> <tr> <th>Criteria</th><th>Marks</th></tr> <tr> <td>• Provides a correct solution</td><td>3</td></tr> <tr> <td>• Demonstrate that k implies k+1</td><td>2</td></tr> <tr> <td>• Demonstrate the first case</td><td>1</td></tr> </table>	Criteria	Marks	• Provides a correct solution	3	• Demonstrate that k implies k+1	2	• Demonstrate the first case	1	
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	Sample solution: when $n = 1$ $\ln\left(\frac{1+1}{1}\right) = \ln(2)$ $= \ln(1+1)$ $\therefore$ statement true for $n = 1$ Assume true for $n = k$ that is: $\ln(2) + \ln\left(\frac{3}{2}\right) + \ln\left(\frac{4}{3}\right) + \dots + \ln\left(\frac{k+1}{k}\right) = \ln(k+1)$ when $n = k+1$ $\ln(2) + \ln\left(\frac{3}{2}\right) + \ln\left(\frac{4}{3}\right) + \dots + \ln\left(\frac{k+1}{k}\right) + \ln\left(\frac{k+1+1}{k+1}\right)$ $= \ln(k+1) + \ln\left(\frac{k+1+1}{k+1}\right)$ $= \ln((k+1)+1)$ as required $\therefore$ as $n=1$ is true and $n=k$ true proves $n=k+1$ is true the statement is true by the principle of mathematical induction. <i>Some students made logically false statements. learn the process of induction</i>									
	ii) Hence find p for which $\sum_{n=1}^p \ln\left(\frac{n+1}{n}\right) \geq \pi$	1								
	<table> <tr> <th>Criteria</th><th>Marks</th></tr> <tr> <td>• Provides a correct solution</td><td>1</td></tr> </table>	Criteria	Marks	• Provides a correct solution	1					
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	Sample solution: $\sum_{n=1}^p \ln\left(\frac{n+1}{n}\right) \geq \pi$ $\ln(p+1) \geq \pi$ $p+1 \geq e^\pi$ $p \geq e^\pi - 1$ $p \geq 22.14\dots$ $\therefore p = 23$ <i>mostly done well</i>									

sum of

14a) (i)	From the equations of motion, $V \cos \theta = \frac{x}{t}$ $V \sin \theta = \frac{y}{t} + \frac{1}{2}gt$ $\left(\frac{x}{t}\right)^2 + \left(\frac{y}{t} + \frac{1}{2}gt\right)^2 = V^2(\cos^2 \theta + \sin^2 \theta)$ $(x)^2 + \left(y + \frac{1}{2}gt^2\right)^2 = V^2t^2$ $x^2 + y^2 + ygt^2 + \frac{1}{4}g^2t^4 = V^2t^2$ $4y^2 + 4ygt^2 + (4x^2 + g^2t^4 - 4V^2t^2) = 0$	2 marks: correct proof 1 mark: substantial progress is made in trying to eliminate $\sin \theta$ and $\cos \theta$	Students who could understand, "eliminate θ was the purpose of the question did it correctly. Well done by many students
(ii)	$t = \frac{x}{V \cos \theta}$ where $x=d$ and V are constants. Hence, t is minimum when $\cos \theta$ is maximum which occurs when $\theta = 0$. i.e. The projectile is fired horizontally. 1 mark $\theta = 0$, then $\cos \theta = 1$. Hence, $t = \frac{x}{V \cos \theta} = \frac{d}{V}$ seconds. 1 mark	1 mark: proves angle of projection $\theta = 0$. 1 mark: proves the result with correct working and reasoning.	Very few students realised that "first" impact means, you need to minimise t. Most of the students <u>used</u> $\theta = 0$ to prove $t = \frac{d}{V}$ and vice versa. These students were awarded 0 marks. They have not proved any of the required results.
(iii)	$y = -\frac{1}{2}gt^2 + Vt \sin \theta$ $\theta = 0$, then $\sin \theta = 0$. $y = -\frac{1}{2}g\left(\frac{d}{V}\right)^2 = -\frac{gd^2}{2V^2}$ (hits below the horizontal.)	1 mark: substitutes $\theta = 0$ into the equation and gives the correct result.	Most students got this mark, except those who did not realise $\theta = 0$.
(iv)	To have two impacts on the wall, there need to prove that here are two solutions for y when $t > \frac{d}{V}$. $4y^2 + 4gt^2y + (g^2t^4 + 4x^2 - 4V^2t^2) = 0$ $x = d$, $\Delta = (4gt^2)^2 - 4 \times 4(g^2t^4 + 4d^2 - 4V^2t^2)$ $= 16g^2t^4 - 16(g^2t^4 + 4d^2 - 4V^2t^2)$ $= 64(V^2t^2 - d^2)$. When $t > \frac{d}{V}$, $Vt > d$. Hence, $V^2t^2 - d^2 > 0$, and $\Delta > 0$. i.e. There are two solutions for y. This means that there are two real and	1 mark: proves $\Delta > 0$ for the quadratic in y.	Many students did not attempt this question. Those who realised two impacts at given t meant that there are two values for y. Majority of the students who attempted this question got 1 mark for setting $\Delta > 0$.

	distinct roots for y and so, there are two impacts at the same t. 1 mark. Now the distance between the impacts equals the difference between the roots. The roots are $\frac{-b \pm \sqrt{\Delta}}{2a}$ Hence the difference = $\frac{2\sqrt{\Delta}}{2a} = \frac{\sqrt{\Delta}}{a}$ $= \frac{\sqrt{64(V^2t^2 - d^2)}}{4}$ $= 2\sqrt{V^2t^2 - d^2} \quad \text{1 mark}$ Another approach: $y_1 + y_2 = -gt^2$ $y_1y_2 = \frac{g^2t^4 + 4x^2 - 4v^2t^2}{4}$ $y_1 + y_2 = \sqrt{(y_1 + y_2)^2 - 4y_1y_2}$ $= \sqrt{g^2t^4 - g^2t^4 - 4x^2 + 4v^2t^2}$ $= 2\sqrt{v^2t^2 - x^2}$ $= 2\sqrt{v^2t^2 - d^2}$	1 mark: correct answer from correct working	
(v)	Distance = $2\sqrt{V^2t^2 - d^2}$ $= 2\sqrt{100t^2 - 100}$ $= 20\sqrt{t^2 - 1}$ $20\sqrt{t^2 - 1} = 20\sqrt{3}$ $t^2 - 1 = 3$ $t = 2, \quad t > 0$ $\cos\theta = \frac{x}{Vt} = \frac{10}{10 \times 2} = \frac{1}{2}$ Hence, $\theta = 60^\circ$ and -60° are the angles of projection.		<i>Students who realised this question is a follow up of (iv) got the mark.</i> <i>Many set $y = 20\sqrt{3}$ in $y = V \sin\theta - \frac{1}{2}gt^2$ and tried to solve for t unsuccessfully.</i>

14b) (i)	$\angle OTN = 90^\circ$ ($OT \perp TN$, angle between tangent and radius is 90°) $\angle OPN = 90^\circ$ (the line from the centre that bisects the chord is perpendicular to the chord) 1 mark $OPNT$ is cyclic as the opposite angles add to 180° 1 mark	1 mark: both reasoning statements are correct 1 mark: gives the reason for the quadrilateral to be cyclic.	<i>Well done.</i> A very common error the statement: Line from the centre to the tangent makes 90°.  Clery the diagram above doesn't. Also, as shown in the answers, only if the chord is bisected, it is perpendicular. Note: you need to learn and present the statements of the theorems accurately.
(ii)	Similarly, $\angle OPM = 90^\circ$ (from (i) $OP \perp PN$ and hence MN.) $\angle OSM = 90^\circ$ (angle between tangent and radius is 90°) $OPSM$ is cyclic as the angles subtended by arc OM in the same segment are equal.) 1 mark	1 mark: gives the correct reasoning for the quadrilateral to be cyclic.	
(iii)	$\angle OTS = \angle OST = \theta$ ($OT = OS$ radii, angles opposite equal sides of isosceles $\triangle OTS$ 1 mark Now, $\angle OTP = \angle ONP = \theta$ and $\angle OSP = \angle OMP = \theta$ angles in the same segment of cyclic quadrilaterals $OTNP$ and $OPSM$. 1 mark Hence, $\angle ONP = \angle OMP = \theta$.	2 marks: both reasoning correct 1 mark only: if substantial progress	A common error was in the interpretation of the diagram. Some students misunderstood the notation on the diagram meant $MP = PN$, rather than $AP = PB$. You always need to verify the diagram from the description of facts stated in the question.
(iv)	$\triangle OMN$ is isosceles ($OM = MN$ as $\angle ONP = \angle OMP = \theta$. sides opposite equal angles are equal) OP is the altitude. Hence, $MP = PN$ (perpendicular bisects the opposite side) Hence, $MP - AP = PN - PB$ (equals subtracted from equals) ie. $AM = BN$	1 mark: correct reasoning	<i>Well done</i>